Python Dictionary

map / Hashing

```
d = {1: 'ABC', 2: 'DEF'}
d[3] = 'xyz'
```

1) { }
2) element → item [key : value]
3) ordered
4) keys( ), values( ), items( )
5) dict( )
6) Keys are unique and im-mutable
7) values → mutable
   → Duplicate

KEYS
(Number, tuple)
storing

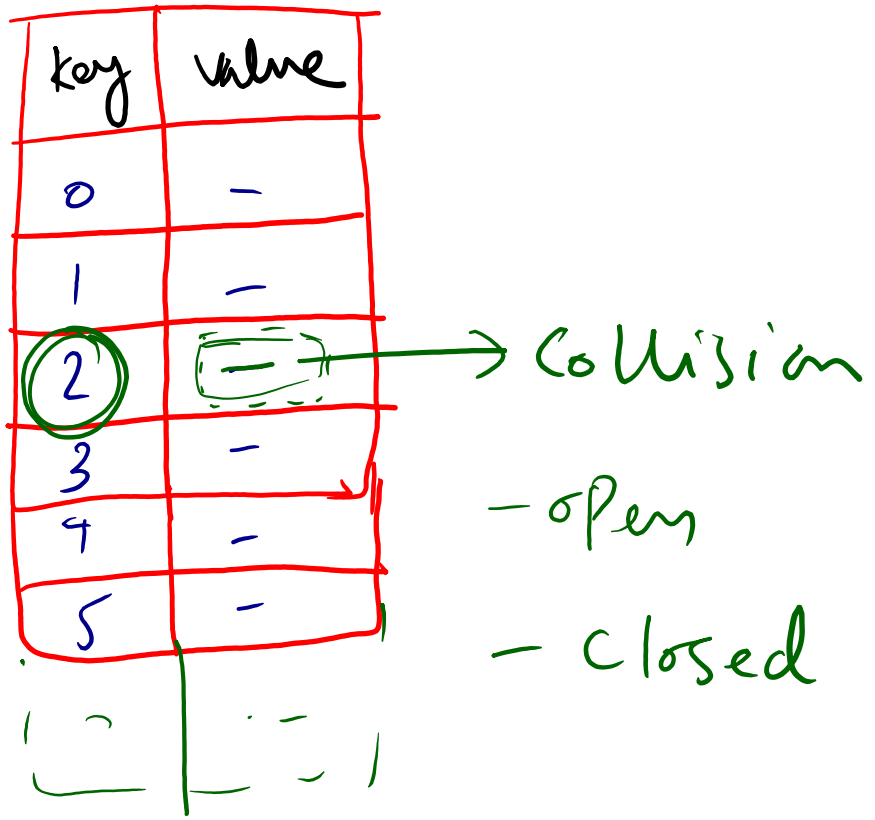
Python set

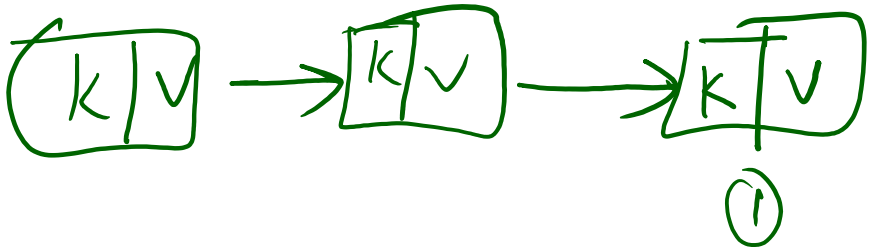
1. Heterogenious
2. Un-ordered [No indexing]
3. unique
4. add( ), pop( )
5. union, Intersection, difference, symmetric_difference
   (|) (&) (-) (^)

Hashing :

1) Hash-function

Pre-Storing values into some data-structure [list | set | dict] and retrieve those values.

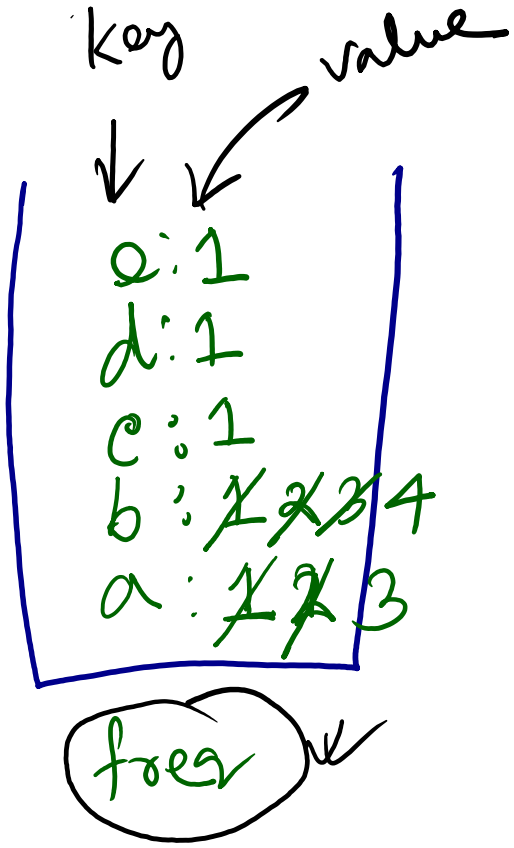




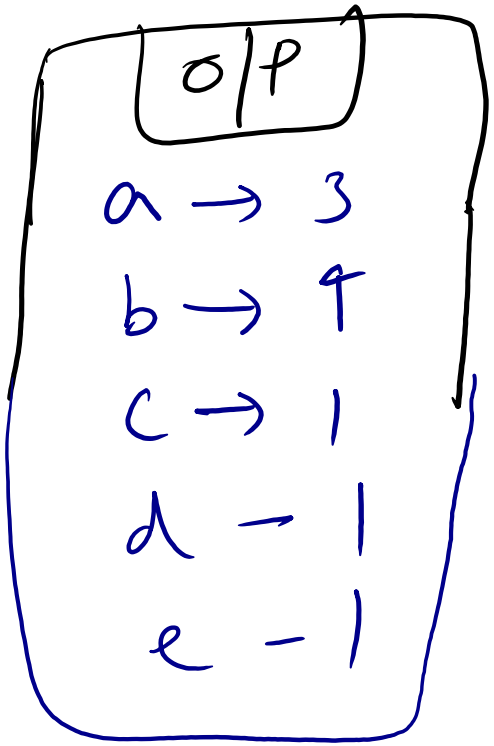
Frequency - dictionary:

freq = { }

S = "abbcadeabb"




```
freq = { }
for i in range(len(S)):
    if S[i] not in freq:
        freq[S[i]] = 1
    else:
        freq[S[i]] += 1
print(freq)
```



Two-Sum

TLE

arr = [2, 3, 8, 5, 7, 1] , target = 15

0 1 2 3 4 5

freq = { }

for i in range(len(arr)):

rem = target - arr[i]

if rem in freq:

return [freq[arr[i]], i]

else:

freq[arr[i]] = i

15 - 2 = 13

15 - 3 = 12

15 - 8 = 7

15 - 5 = 10

15 - 7 = 8

[2, 4]

Ans

5 : 3

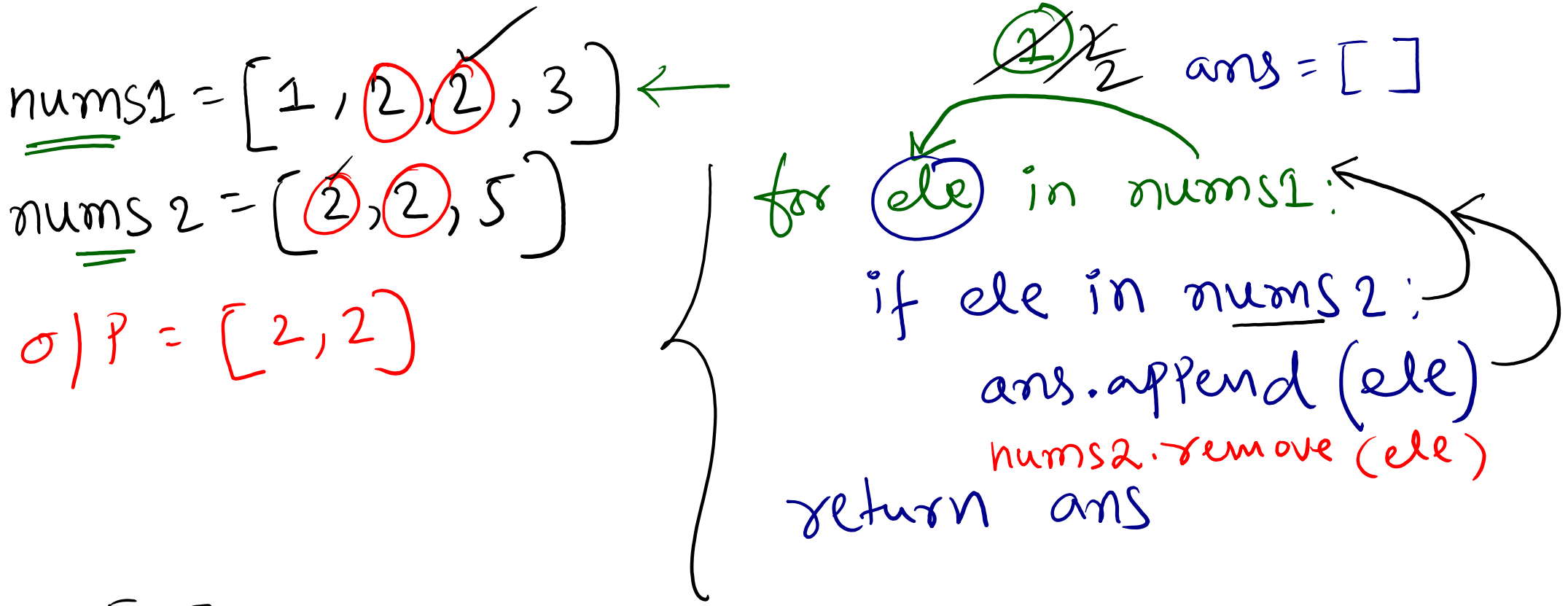
8 : 2

3 : 1

2 : 0

freq[rem]

freq



ans = [2]

ans = [2,2] → o/p